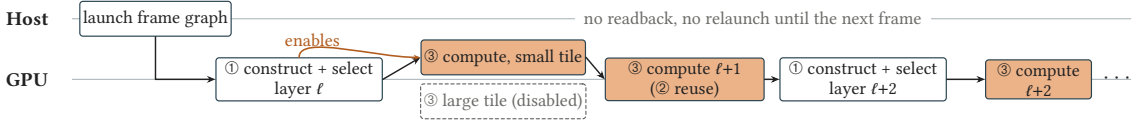

① Construction + selection
tile 0
tile 1
tile 2
tile 3
$M_{out}$
none
$\mathcal{W}$
length $n_{\mathcal{W}}$
small tile
large tile
per-layer thresholds, chosen offline
② Reuse
$\mathcal{W}$
conv $\ell$
conv $\ell$+1
new geometry
$\mathcal{W}'$ by ①
conv, stride 2
③ Compute
$\mathcal{W}$
tile 0
tile 1
tile 2
loop over $\mathcal{W}$
SM 0
SM 1
$\mathcal{N}(9)$
$\mathcal{N}(12)$
$\mathcal{N}(14)$
$\mathcal{N}(0)$
$\mathcal{N}(5)$
$\otimes$ f =
$y_9$
$y_{12}$
$y_{14}$
$y_0$
$y_5$
tile 0
tile 2
tile 1
$\mathcal{N}(p)$: read from x in place;
$\otimes$ f: the dense reduction


Host
launch frame graph
no readback, no relaunch until the next frame
GPU
enables
① construct + select layer $\ell$
③ compute, small tile
③ large tile (disabled)
③ compute $\ell$+1 (② reuse)
① construct + select layer $\ell$+2
③ compute $\ell$+2